Original Manuscript ID: Access-2026-35630

Original Article Title: "Design and Simulation of a Broadband Tunable Transmittance Optical Filter Using Sb2Se3 Phase Change Material "

To: IEEE Access Editor

Re: Response to reviewers

Dear Editor,

Thank you for allowing a resubmission of our manuscript, with an opportunity to address the reviewers' comments.

We are uploading (a) our point-by-point response to the comments (below), (b) an updated manuscript with yellow highlighting indicating changes (as "Highlighted PDF"), and (c) a clean updated manuscript without highlights ("Main Manuscript"*).*

Best regards,

Surawut Wicharn, et al.

Reviewer#1, Concern # 1 (Clarify the novel and innovative aspects of the work by comparing the proposed design with state-of-the-art Sb_2Se_3-based optical filters.):

Author response: We add a table comparing our 5-unit-cell long-pass edge-filter design to other PCM-based optical filter designs at the IV.C RESULTS & DISCUSSIONS.

Author action:

Reviewer#1, Concern # 2 (Support simulation results with experimental evidence or published experimental data.):

Author response: Since we don't have fabrication capabilities, a lack of experimental validation has been added in the IV.D. Limitation and Future Works Suggestion section.

Author action:

Reviewer#1, Concern # 3 (Provide a stronger justification for the simulation parameters and assumptions employed.):

Author response: This comment is very broad so we couldn't provide a direct response. But we have already stated every simulation parameters: simulation interval (every 1 nm), source of complex refractive index ($\tilde{n}$) dataset, interpolation method for missing datapoint, thickness of each layer, thickness of SiO_2 substrate, incident angle, polarization modes.

Author action: We also add the interpolation method for missing datapoint "Missing datapoint are linearly interpolated since every dataset used have measuring interval of less than 4 nm." in III. METHODS section, previously. Previously, it was only written in II.A. OPTICAL PROPERTIES OF ANTIMONY TRISELENIDE.

Reviewer#1, Concern # 4 (Include a comparison table featuring currently available state-of-the-art optical filters.):

Author response:

Author action: We updated the manuscript by ….

Reviewer#1, Concern # 5 (Validate the endurance model using experimental data or stronger evidence from the scientific literature.):

Author response: This is similar to Concern #2. We already provided the number for the endurance parameters, which are directly taken from XXX (the measurement is from single layer bulk Sb_2Se_3). (พี่เขม)

Author action: We mention the lack of experimental validation in the Limitations and Future Work Suggestions section.

Reviewer#1, Concern # 6 (Discuss the feasibility of practical fabrication and implementation challenges.):

Author response: (เขม)

Author action: We updated the manuscript by ….

Reviewer#1, Concern # 7 (Assess the impact of fabrication tolerances on device performance.):

Author response: This is also a common concern among other reviewer, so we add "Simulation of fabrication and refractive index tolerances". We add another simulated the transmittance contrast of the 4- and 5-unit-cell designs by varying each layer thickness by ±5% and the complex refractive index value of each material by ±2%.

Author action: We updated the manuscript by ….

Reviewer#1, Concern # 8 (Strengthen the discussion section by providing a deeper analysis of the simulation results.):

Author response: The discussions and analysis have already been provided in detail directly after each simulated result.

Author action: We think that the existing discussions are sufficient, so we haven't taken any specific action to this comment.

Reviewer#1, Concern # 9 (Expand the conclusion section to include limitations and future research directions.):

Author response: The current limitations and future work suggestions have been added to the IV. Results & Discussion and V. Conclusion.

Author action: We updated the manuscript by ….

We thank Reviewer#1 for your valuable suggestions.

Reviewer#2, Concern # 1 (Clarify the novelty by explicitly distinguishing the proposed symbol-level constant-envelope waveform design from existing constant-envelope OTFS and CPM-based ISAC approaches in the Introduction and Related Work.):

Author response: To the best of our knowledge, "Constant-envelope waveform design" has no relation to any designs in this work. However, we have added the comparison between our design and previously published design (Sb_2Se_3-based and other PCM-based).

Author action: We didn't take any action to this particular concern.

Reviewer#2, Concern # 2 (The entire study is TMM simulation; there is no fabricated sample, measured spectrum, or comparison to any prior experimental Sb_2Se_3 filter to anchor the modeled numbers (transmittance, ΔT, endurance). Given IEEE Access's broad but rigor-focused scope, at minimum a sensitivity/tolerance analysis (e.g., ±5–10% layer thickness error) would strengthen practical credibility.):

Author response: Simulation of thickness tolerance (±5%) and refractive index tolerance (±2%) has been added to section XXX. We have also added a comparison table to other PCM-based optical filters.

Author action:

Reviewer#2, Concern # 3 (The paper repeatedly claims benefits like "less Sb_2Se_3 used... minimizes energy required for phase transition" (Sec. IV.A) and discusses electrical vs. optical heating for the multilayer configurations (Sec. 4.3), but no switching energy, power, or thermal simulation is provided to substantiate these practical tunability claims.):

Author response:

Author action:

Reviewer#2, Concern # 4 (λ_0 = 740 nm is chosen "as an example" to align the AR/HR edge with 1064 nm, and only afterward is an optimizer applied (Sec. 4.3). Explaining why 740 nm specifically (vs. a systematically swept λ_0) would improve rigor; the current framing suggests trial-and-error justified post hoc.):

Author response: We have added Fig. 5 showing the reflectance profile of four-unit-cell edge-filter designs with reference wavelength (λ_0) varying from 700 to 750 nm. We didn't add this figure before since we thought it would confuse the reader.

Author action:

Reviewer#2, Concern # 5 (Fig. 7 compares only N=4 and N=5 edge filters; the final chosen design is N=5, but no rationale is given for omitting N=3 or N=6 from this specific comparison, especially since N=6 has the highest ΔT reported in Table I.):

Author response:

Author action: The missing reason for selecting 5-unit-cell design has been added at the end of section IV.A. A. OPTICAL PROPERTIES OF 1064-NM OPTICAL FILTERS: “XXX”

Reviewer#2, Concern # 6 (Improve reproducibility by providing additional simulation details (e.g., parameter settings, initialization, convergence criteria) and indicate whether the simulation code will be publicly available.):

Author response: Same as Reviewer #1, Concern #3. All parameters setting for the optical simulations have already been provided in the III. Methods section (simulation spectral range and interval, refractive index dataset of each layer and substrate, substrate thickness, incident angle and polarization mode. The layer thickness is not specifically mentioned in nm but as QWOT where the reference wavelength has also been provided. We did this because the specific thickness of each layer depends on its complex refractive index, which can differ.

The initialization and convergence criteria for optimization were also already provided in the IV.C. Transmittance tunability and optimization.

The simulation code is uploaded to private GitHub and will be shared upon a reasonable request.

Author action:
We updated the manuscript by updating the Data Availability section to mention the simulation codes “Data and simulation codes underlying the results presented in this paper may be obtained from the corresponding author upon a reasonable request.”

Reviewer#2, Concern # 7 (Table I column header "SbSe used (nm)" should read "Sb_2Se_3 used (nm)"; also clarify FWHM computation method (measured on the ΔT spectrum vs. individual Ta/Tc curves) since this metric is quoted precisely (197.0 nm, 207.4 nm) without defined methodology. Minor presentation improvements are recommended, including proofreading for notation consistency, ensuring all variables are defined upon first use, and improving the readability of several equations and figures.):

Author response: The correction of Table I header is appreciated. The FWHM is measured from ΔT spectrum. This was previously written as ‘the FWHM of high-ΔT region is 197.0 nm’, the ‘high-ΔT’ might be the cause of this confusion. So, we remove the ‘high-‘ part. We have also added this in the III. Methods section

Author action: We have updated Table I header to the suggested. The FWHM of *ΔT* spectrum “the high-contrast (*ΔT*) region spectral width” in the III.A. SIMULATION OF OPTICAL PERFORMANCES section have been replaced with “the transmittance contrast ΔT FWHM”.

Thank you Reviewer #2 for the comments.

Reviewer#3, Concern # 1 (This work is simulation-only. If possible, include experimental validation; otherwise, discuss fabrication feasibility.):

Author response: This is the same as Reviewer#1 Concern #2 and Reviewer#2 Concern #2. Since experimental validation is impossible due to our lack of equipment, we added this limitation to the Limitation and Future Work section. However, we have written the fabrication feasibility to the latter part of IV. RESULTS & DISCUSSIONS. พี่เขม

Author action:

Reviewer#3, Concern # 2 (Add a benchmark comparison with existing tunable filters at the end of Section IV to better highlight the advantages of this design.):

Author response: We add the benchmark table to IV.D. Comparison to Other PCM-based Optical Filters.

Author action:

Reviewer#3, Concern # 3 (Figure 7 shows that beyond 30° incidence, contrast degrades significantly for both polarizations. Specify 0°–30° as the effective working range and briefly explain the physical cause.):

Author response: We have specified the working range of the 5-unit-cell design as commented. The cause is due to the ΔT blueshift with increasing incident angle. The reason is surprisingly counterintuitive, since increasing angle increase effective thickness of each layer which intuitively should've been equivalent to increasing λ_0. But since

This is a very great point, and we thank the reviewer for mentioning this.

Author action: We updated the manuscript by describing the text of Fig.7 (now Fig.8) in more details: "

Reviewer#3, Concern # 4 (Add a schematic diagram of the interference mechanism in Section II to improve readability.):

Author response: The structural architecture of two-layer, three-layer, and long-pass edge-filter designs has been added to Fig. XX

Author action:

Reviewer#3, Concern # 5 (Unify the legend styles across all subfigures in Figure 8.):

Author response: พี่เขม: ให้ขนาด block legends ในรูปเหมือนกันหมด (ตามความเข้าใจผม + อจ.). เพิ่มขนาด แกน X, Y

Author action:

Reviewer#3, Concern # 6 (Equations (14) and (15) are repetitive; combine or simplify them.):

Author response: We did separate the two equations to avoid confusion for new reader but has since combined them per reviewer suggestion.

Author action: We updated the manuscript by combining Eq. 14 and 15 to the new Eq. 14. And updated every following mentioned of “Eq.” in the text to the corrected one.

Thank you for your valuable suggestions. It’s greatly appreciated.

Reviewer#4, Concern # 1 (The paper claims a broadband tunable optical filter, yet the authors manually assign different phase configurations inside the simulation, which makes the idea of continuous tuning and wavelength tuning difficult to experimentally realize. The author demonstrates many possible optical states but not necessarily a realizable tunable optical filter.):

Author response: We have never claimed that our work is continuously tunable, the transmittance value tunability is shown in steps by changing each layer phase as illustrated in Fig. 9, as mentioned by the Reviewer.

The detailed phase switching methods or thermal simulation of each Sb_2Se_3 layer were not provided in this work. This is now highlighted in the Limitations and Future Works section.

Author action: The lack of switching methods or thermal simulations in this work are now written in IV.E. LIMITATIONS AND FUTURE WORKS SUGGESTIONS.

Reviewer#4, Concern # 2 (This paper mostly changes transmission amplitude instead of substantially shifting the filter spectrum. The optimization only slightly modifies the spectrum (~ < 50 nm). That is optimization, not significant wavelength tunability.):

Author response: We think the Reviewer misunderstand the meaning of tunability in this work. Transmittance tunability is the ability the change the transmittance values by switching the phase of Sb_2Se_3 which does not specifically mean the tunability of wavelength.

The idea that the reviewer is confused is supported by the fact the reviewer mentioned ~ < 50 nm modified spectrum due to optimization. The optimization goal has never been to shift the operating wavelength, but the goals are for broad and stable transmittance contrast, shown in Fig. 10b (now Fig. 11b). But as seen in Fig. 5, the edge between AR and high-reflection regions can be shifted by ~200 nm by changing every Sb_2Se_3 layer from amorphous to crystalline phase.

Author action: As a response, we have not taken any direct action due to this concern.

Reviewer#4, Concern # 3 (Tables I and II optimize layer thicknesses on the ~nm scale using classical thin-film interference. The paper assumes bulk refractive index and extinction coefficient are independent of thickness. The use of bulk values should be justified for those that could have arisen from finite-thickness effects, interface quality, and crystallization non-uniformity.):

Author response: ดูงานอื่นว่าพูดถึง finite effect value หรือไม่

We haven't thought about finite-thickness effects since recent work with even smaller Sb_2Se_3 layer does not concern about finite-thickness effects either [1]. However, we think that this is an interesting points and will be mentioned this in the Future Work Suggestions section.

We are going to partially rectify the concern about using thickness-dependent complex refractive index in this work by simulating the original and optimized 5-unit-cell designs by varying the thickness of each layer by ±5% and vary each material refractive index by ±2%.

We would like to explain that the detailed simulation for the exact optical performance values is outside the scope of this work. The main point of our work is to highlight that Sb_2Se_3 is suitable for tunable transmittance optical filter with a near-zero transmission-loss, not to directly compare the design performances.

Author action: We add sentences about finite-thickness effects

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Reviewer#4, Concern # 4 (Fig. 8 demonstrates propagation of assumed material degradation. The endurance simulation follows simple exponential and linear drift equations, whose coefficients are taken from literature, rather than derived from phase-transition simulations.):

Author response: พี่เขม

Author action: We updated the manuscript by ….

Reviewer#4, Concern # 5 (For 93.6% transmission, the insertion loss is ~ IL = -10 log10(0.936) ~ 0.3 dB. However, Fig. 8 reports insertion loss evolution relative to the initial state, not absolute insertion loss. The authors should also report: a) initial IL, b) final IL, and c) absolute IL. Also, for non-monotonic behavior observed in 4 layers => 5 => 6, no systematic explanation is given. Justification given through field confinement and resonance is not explained properly for this non-monotonic behavior.):

Author response: พี่เขม

Author action: We updated the manuscript by ….

Reviewer#4, Concern # 6 (Motivation of the presented work is missing in the introduction and abstract. The authors state Sb2Se3 has not been explored much, but the paper cites recent Sb2Se3 optical filter studies and does not quantitatively compare against them. How does this paper stand out from recent work?

1) https://iopscience.iop.org/article/10.1088/2040-8986/ad80a6/meta;
2) https://opg.optica.org/ome/fulltext.cfm?uri=ome-12-11-4268;
3) https://opg.optica.org/oe/fulltext.cfm?uri=oe-34-5-7378

Why another filter? What gap exists? What limitation exists? What application cannot currently be achieved? A comparison table against the works needed to strengthen the novelty claim and strengthen the introduction.):

Author response: We couldn't access the first provided paper. The second paper is . We have created a benchmark table (Table II.) at the end of IV. RESULTS & DISCUSSIONS. Our design has a better transmission in ON mode, higher optical contrast, can be operated for broader wavelength but require more layer and the device size is thicker. It should be noted that this is not a direct comparison since the design is for different wavelengths (1064 vs XXX nm).

Author action: We updated the manuscript to include the comparison table.

Thank you for the detailed responses and interesting points.

Reviewer#5, Concern # 1 (Q1: In the abstract,

1) please consider adding one sentence on why a switchable filter at this wavelength is needed and what limits existing PCM options.

2)if the work is purely simulation-based, use language like "simulation results show" rather than definitive statements, to accurately reflect the study's nature.):

Author response: We have added a brief introduction to the abstract and change the wording to "simulated result".

Author action:

Reviewer#5, Concern # 2 (Q2: The Introduction spends considerable space defining PCM and its classifications (volatile vs. non-volatile, VO2's transition mechanism, etc.), which may be more detail than necessary for this journal's readership. More importantly, I could not find a clear, explicit statement of what specific problem or limitation in existing PCM-based tunable filters this work aims to solve—could the authors clarify what practical shortcomings of current materials/designs (e.g., excessive loss, low endurance, narrow bandwidth) motivated this study, and state this more directly before introducing their proposed approach?):

Author response:

Author action:

Reviewer#5, Concern # 3 (Q3: The drift coefficient (α_p) and loss growth coefficient (β_p) in Equations 18–19 are stated to be "consistent with experimental trends," but no specific numerical values or their literature source are given. Could the authors clarify exactly which experimental dataset these coefficients were fitted to, and report the actual values used in the simulation?):

Author response: พี่เขม

Author action: We updated the manuscript by ….

Reviewer#5, Concern # 4 (Q4: The paper states the 5-unit-cell design offers "the best compromise" between performance, size, and complexity, but this appears to be a qualitative judgment. Could the authors provide a quantitative figure-of-merit (e.g., a weighted score combining ΔT, thickness, and material usage) to justify why N=5 was chosen over N=4 or N=6, rather than N=6 which showed the highest ΔT?):

Author response: In our opinion, the quantitative value from any figure-of-merit can be weight in such a way to make any *N* be the best design, and there exists no standardized FOM to base on. However, we agree that a stronger reasoning for choosing *N*=5 over other designs is needed, so we give a more detailed qualitative explanation in each relevant paragraph in IV.A. A. OPTICAL PROPERTIES OF 1064-NM OPTICAL FILTERS.

Author action:

Reviewer#5, Concern # 5 (Q6: Since all data are simulation-based and rely on literature-sourced refractive index values (which themselves carry measurement uncertainty), could the authors comment on how sensitive the reported ΔT, FWHM, and endurance values are to small variations (e.g., ±1-2%) in the input n/k datasets?):

Author response: We have since added simulation of thickness and refractive index uncertainty for the four-layer and five-layer unit-cell designs, ±5% and ±2%, respectively.

Author action: We updated the manuscript by ….

Reviewer#5, Concern # 6 (Q7: On the incident angle/polarization data (Fig. 7), The paper notes TM polarization causes a "significant blueshift" at higher incident angles but does not quantify this shift numerically. Could the authors report the specific magnitude of the blueshift (in nm) at, say, 45° and 60° for both the 4- and 5-unit-cell designs?):

Author response: We added the mentioned numerical value to the text before Fig. 7 (now Fig. 8). We have also added a brief explanation of the blue-shifting (Reviewer#3 Concern #3). This figure has also been enlarged to the same width as other figures.

Author action: We updated the manuscript by adding "At higher incident angle, the high-ΔT spectral profile (yellow-green) shifted to shorter wavelengths. For TE polarized light, the blueshift of wavelength with the highest ΔT at 45° compared to 0° incident angle are -37 and -57 nm for 4-unit-cell and 5-unit-cell designs, respectively. And at 60° angles the blueshift are -90, -143 nm, respectively. As for TM polarized light, the peak ΔT blueshift in 4-unit-cell and 5-unit-cell designs at 45° are -103 and -168 nm, respectively. At 60°

incident angles they are -156 and -226 nm, respectively. It can be seen that the ΔT profile of both polarization modes blue shifted with increasing incident angle, with TM polarization causing longer blueshift.”. The corresponding Fig. 7 (now Fig. 8) has also been enlarged to full-page size.

Note: *References suggested by reviewers should only be added if it is relevant to the article and makes it more complete. Excessive cases of recommending non-relevant articles should be reported to ieeeaccesseic@ieee.org*

[1] S. Blundell *et al.*, "Ultracompact Programmable Silicon Photonics Using Layers of Low-Loss Phase-Change Material Sb2Se3 of Increasing Thickness," *ACS Photonics,* vol. 12, no. 3, pp. 1382–1391, 2025/03/19 2025, doi: 10.1021/acsphotonics.4c01789.